

EXCEL TRAINING SESSIONS

- REFRESHER INTERMEDIATE

Relative, absolute and mixed references

- Use of relative references in basic sum formula:
In the 'Rel+Abs' sheet, calculate the total amount per row for the three months
- Use the absolute reference of the VAT amount to calculate the total with tax.
- Use a defined name to calculate the total monthly sales with a 10% tax and use a mixed reference in the formula so that you can copy that column to calculate the total with a 12% tax amount.

jan	feb	mar	total	Tax8%	Tax10%	Tax12%
100	200	300	600	648 CHF	660 CHF	672 CHF
110	205	302	617	666 CHF	679 CHF	691 CHF
120	210	304	634	685 CHF	697 CHF	710 CHF
130	215	306	651	703 CHF	716 CHF	729 CHF
140	220	308	668	721 CHF	735 CHF	748 CHF
150	225	310	685	740 CHF	754 CHF	767 CHF
160	230	312	702	758 CHF	772 CHF	786 CHF
170	235	314	719	777 CHF	791 CHF	805 CHF
180	240	316	736	795 CHF	810 CHF	824 CHF
190	245	318	753	813 CHF	828 CHF	843 CHF
200	250	320	770	832 CHF	847 CHF	862 CHF
210	255	322	787	850 CHF	866 CHF	881 CHF
220	260	324	804	868 CHF	884 CHF	900 CHF
230	265	326	821	887 CHF	903 CHF	920 CHF
240	270	328	838	905 CHF	922 CHF	939 CHF

Mixed references

- In the 'Mixed' Sheet: calculate the various margin amounts for all products by copying the same formula in the last 3 columns.
- Make use of mixed cell references so that the total amounts are correct on every row and every column.

jan	feb	mar	total	Tax8%	Tax10%	Tax12%
100	200	300	600	648 CHF	660 CHF	672 CHF
110	205	302	617	666 CHF	679 CHF	691 CHF
120	210	304	634	685 CHF	697 CHF	710 CHF
130	215	306	651	703 CHF	716 CHF	729 CHF
140	220	308	668	721 CHF	735 CHF	748 CHF
150	225	310	685	740 CHF	754 CHF	767 CHF
160	230	312	702	758 CHF	772 CHF	786 CHF
170	235	314	719	777 CHF	791 CHF	805 CHF
180	240	316	736	795 CHF	810 CHF	824 CHF
190	245	318	753	813 CHF	828 CHF	843 CHF
200	250	320	770	832 CHF	847 CHF	862 CHF
210	255	322	787	850 CHF	866 CHF	881 CHF
220	260	324	804	868 CHF	884 CHF	900 CHF
230	265	326	821	887 CHF	903 CHF	920 CHF
240	270	328	838	905 CHF	922 CHF	939 CHF

Conditional formatting (basic)

- 1st group: Create highlighting rules.

Month	Western	Central	Eastern
1	214 030	103 832	225 732
2	207 192	103 499	230 115
3	199 813	108 260	217 872
4	194 046	111 562	230 386
5	198 252	116 167	223 381
6	208 876	117 365	232 123
7	202 443	119 041	236 201
8	195 685	123 828	231 477
9	189 147	119 624	223 567
10	192 072	117 129	223 352
11	197 941	116 589	223 679
12	202 041	121 501	210 269

Cell value above average

Top 3 values in column

Cell value greater than a given number

Conditional formatting (basic)

- 2nd group:
3 options: Color scales, Data Bars and Icon sets.

Month	Western	Central	Eastern
1	214 030	103 832	225 732
2	218 187	96 977	221 106
3	210 018	93 152	221 803
4	217 361	89 960	220 242
5	208 565	93 365	227 041
6	211 613	97 323	221 655
7	206 619	95 520	230 601
8	213 898	90 052	233 955
9	209 799	94 838	246 233
10	219 324	98 796	257 787
11	213 881	93 991	250 517
12	204 833	95 186	265 199

Row Labels	AVG Length
Tempest	00:06:51
Time Out of Mind	00:06:37
Desire	00:06:14
MTV Unplugged	00:05:50
Modern Times	00:05:49
Highway 61 Revisited	00:05:44
Street Legal	00:05:35
Infidels	00:05:16
Blonde On Blonde	00:05:14
Slow Train Coming	00:05:11
Blood On The Tracks	00:05:11
Tell Tale Signs (Disc 1)	00:05:10
30th Anniversary Concert	00:05:07
Tell Tale Signs (Disc 2)	00:05:00

Month	Western	Central	Eastern
1	➡ 214 030	🌑 103 832	✅ 225 732
2	➡ 209 617	🌑 107 691	✅ 222 538
3	⬇ 207 676	🌑 110 474	✅ 227 561
4	➡ 218 202	🌑 106 601	⚠ 214 603
5	⬆ 224 906	🌑 103 929	❌ 207 302
6	➡ 218 355	🌑 98 888	⚠ 212 516
7	➡ 215 549	🌑 101 270	❌ 203 408
8	➡ 210 355	🌑 95 843	❌ 207 702
9	➡ 220 273	🌑 95 432	❌ 210 994
10	➡ 218 065	🌑 95 363	⚠ 215 150
11	➡ 213 279	🌑 96 115	✅ 227 800
12	⬇ 204 434	🌑 91 035	⚠ 214 233

Conditional formatting (formula)

Name:	Noel					
Alice	7	118	61	55	85	26
Kathy	131	43	197	82	103	163
Laurent	139	30	171	122	34	196
Marie	31	171	185	162	171	17
Noel	78	126	190	78	123	2
Oliver	157	98	100	75	137	10
Patrick	120	144	106	39	39	119
Cléa	156	200	58	74	37	76
Raoul	58	147	160	182	11	79
Sheila	79	183	5	161	104	23
Tom	91	54	100	174	198	78
Ursula	53	140	188	58	54	36
Vincent	121	13	2	139	148	101
Walter	132	65	123	129	174	90
Bertrand	162	127	86	51	164	35
Yolanda	60	116	107	117	189	200
Jean	103	142	103	165	89	37

=A16=\$B\$14
FALSE

685	436	665	565
886	910	699	666
718	56	928	741
904	143	57	849
549	992	455	534
632	578	852	424
871	87	844	209
58	703	693	698
428	493	202	883
350	869	889	133
836	953	171	410
685	208	349	572
898	602	187	277
136	453	571	162
209	864	94	44
439	41	208	600
859	991	527	673
583	192	406	158

=MOD(ROW();2)=0
FALSE

464	381	358	20
236	383	267	242
495	299	473	410
94	199	2	400
152	415	319	141
147	405	222	196
139	344	491	69
152	245	225	336
499	487	22	463
258	285	445	21
82	4	168	253
17	301	4	419
418	146	222	261
378	206	39	317
29	61	417	160
486	1	438	277
296	413	222	372
380	227	114	327
82	137	267	48
325	63	439	74
89	400	233	2

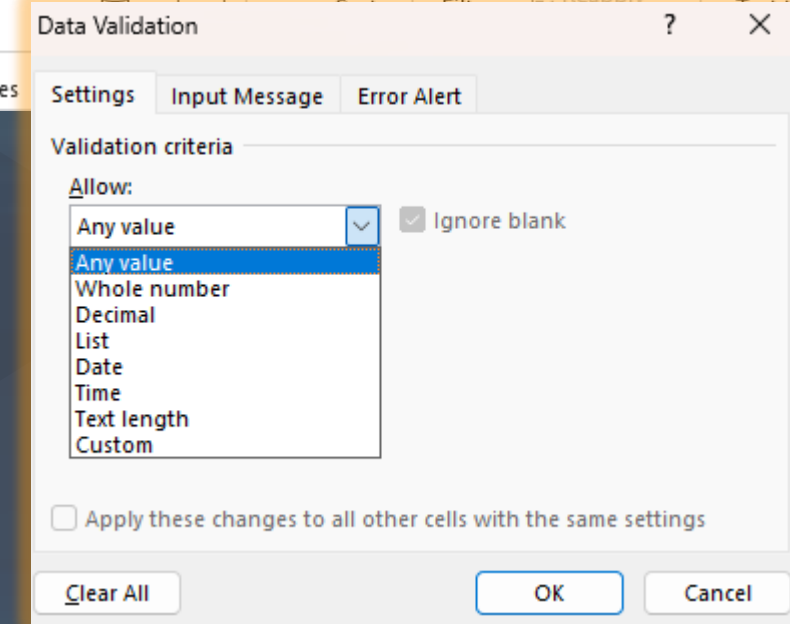
=MOD(INT((ROW()-1)/4)+1;2)=1

	Oct	Nov	Dec
Alice	7	118	61
Kathy	131	43	197
Ursula	139	30	171
Vincent	31	171	185
Walter	78	126	190
Bertrand	157	98	100
Yolanda	120	144	106
Jean	156	200	58

=OR(\$V16="Bertrand";W\$15="Nov")
FALSE

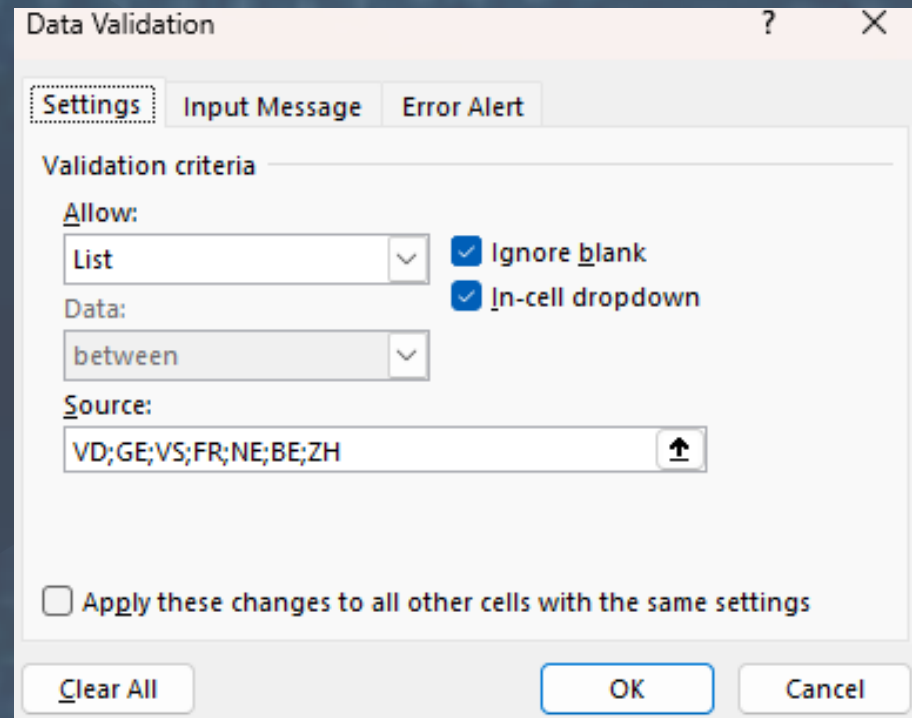
Data validation

- You can force the content of a range to be whole (or decimal) numbers only.
- You can force a date (or time) interval.
- You can force the length of the text in each cell to a certain number of characters.



Data validation: Dropdown list

- You can impose a list of choices in the desired cells.
- The list of choices can come from:
 - ❖ A manual list
 - ❖ A range inside a sheet
 - ❖ A defined name
 - ❖ A formula



The screenshot shows the 'Data Validation' dialog box with the 'Settings' tab selected. The 'Validation criteria' section is expanded, showing the 'Allow:' dropdown set to 'List', the 'Data:' dropdown set to 'between', and the 'Source:' field containing the text 'VD;GE;VS;FR;NE;BE;ZH'. The 'Ignore blank' and 'In-cell dropdown' checkboxes are both checked. At the bottom, there is an unchecked checkbox for 'Apply these changes to all other cells with the same settings' and three buttons: 'Clear All', 'OK', and 'Cancel'.

Data Validation

Settings Input Message Error Alert

Validation criteria

Allow:
List

Data:
between

Source:
VD;GE;VS;FR;NE;BE;ZH

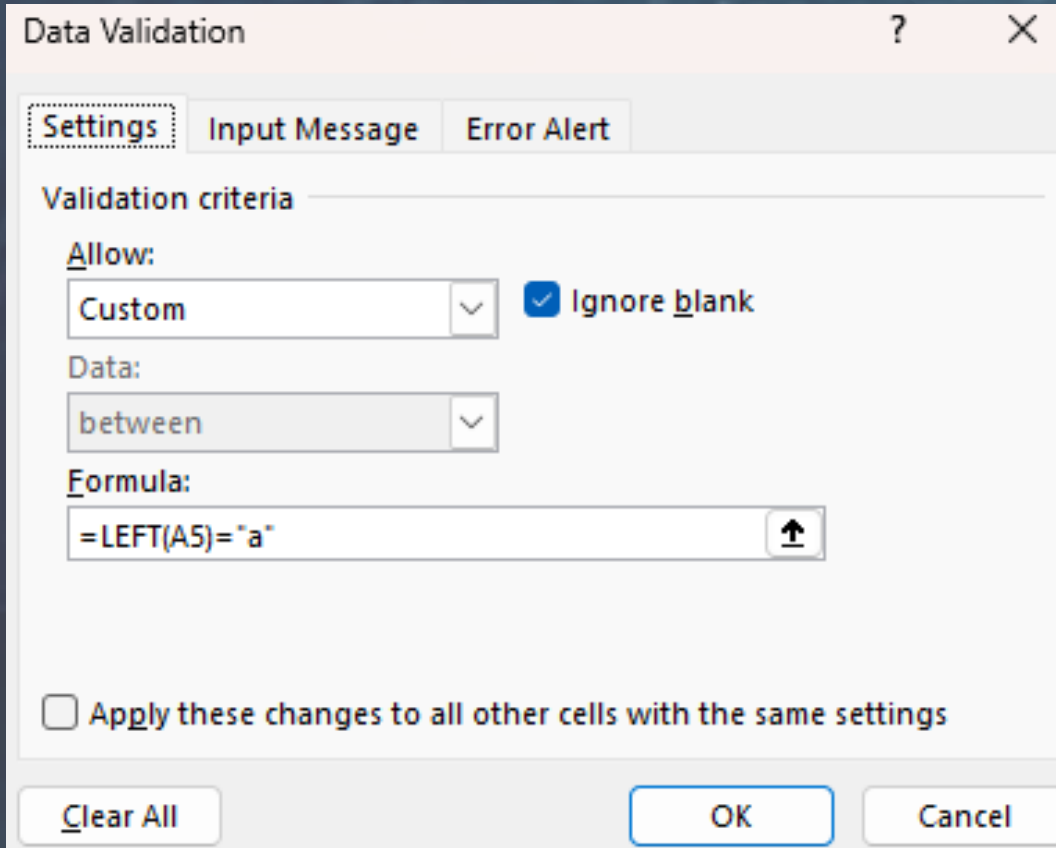
☒ Ignore blank
☒ In-cell dropdown

☐ Apply these changes to all other cells with the same settings

Clear All OK Cancel

Data validation: formulas

- You can customize data validation with formulas



The image shows the 'Data Validation' dialog box in Microsoft Excel. The 'Settings' tab is selected, showing the 'Validation criteria' section. The 'Allow:' dropdown is set to 'Custom', and the 'Ignore blank' checkbox is checked. The 'Data:' dropdown is set to 'between'. The 'Formula:' text box contains the formula '=LEFT(A5)="a"', and the 'Apply these changes to all other cells with the same settings' checkbox is unchecked. The dialog box has buttons for 'Clear All', 'OK', and 'Cancel' at the bottom.

Data Validation

Settings Input Message Error Alert

Validation criteria

Allow:

Custom ☒ Ignore blank

Data:

between

Formula:

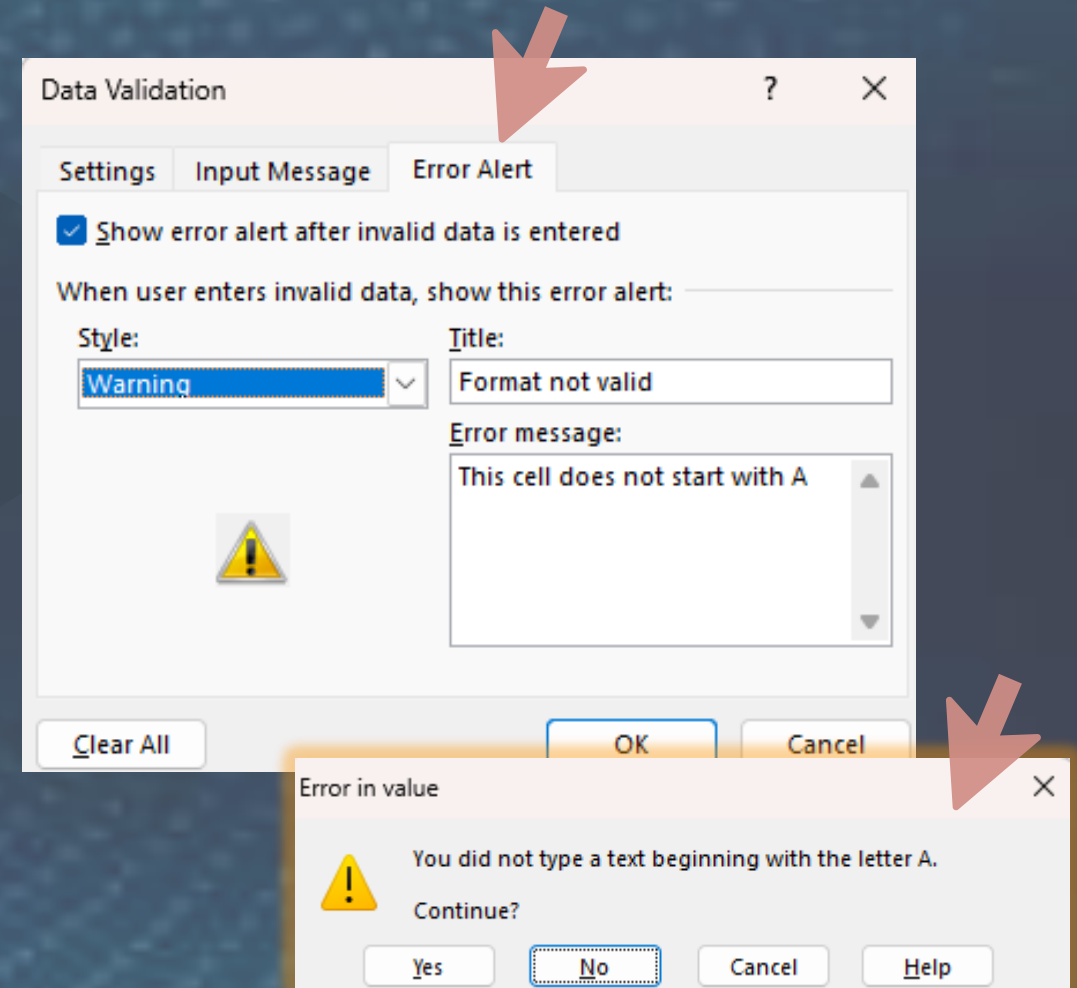
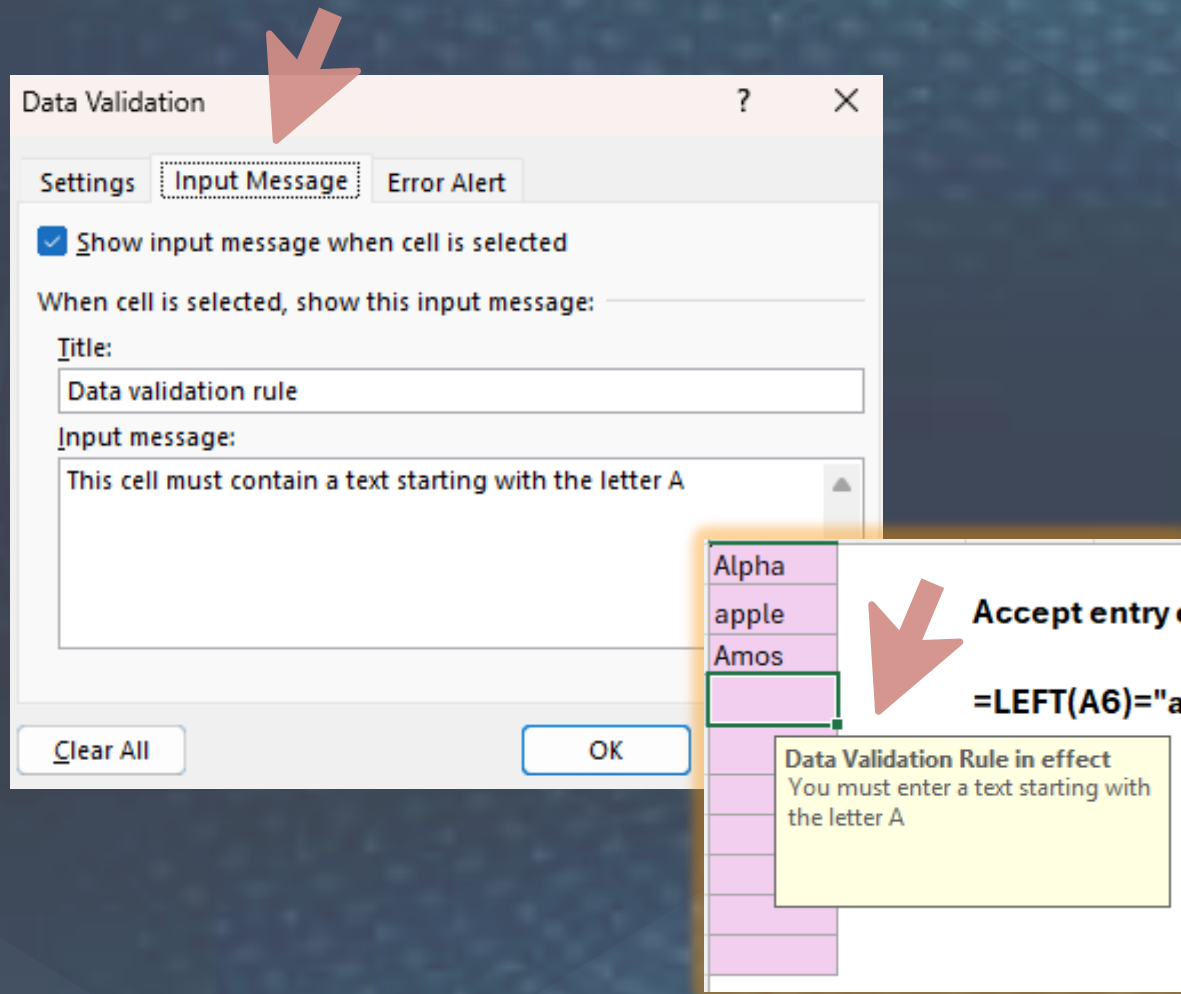
=LEFT(A5)="a"

☐ Apply these changes to all other cells with the same settings

Clear All OK Cancel

Data validation: customize the messages

- You can customize the data validation message when the user .



3D formulas

- Apply modifications or formulas to more than one sheet.
- Use the Sheets interval syntax i.e. SUM(Sheet3:Sheet7!\$D\$12)

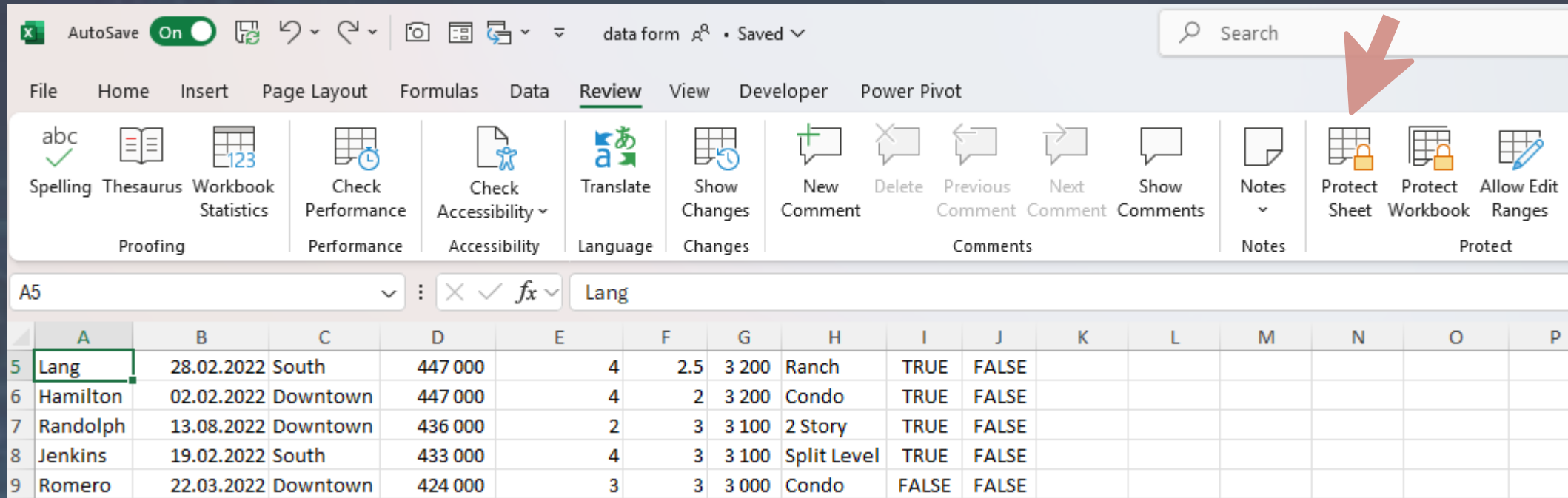
Sem 33	TOTAL HEURES
Lundi	24
Mardi	28
Mercredi	32
Jeudi	36
Vendredi	36
	156

156 =SUM(Décompte_debut:Décompte_fin!B12:B16)

Employés concernés
 Nicole Blanc
 Jean Dubois
 Marcel Devaud
 Robert Borloz
 Anne Velosa

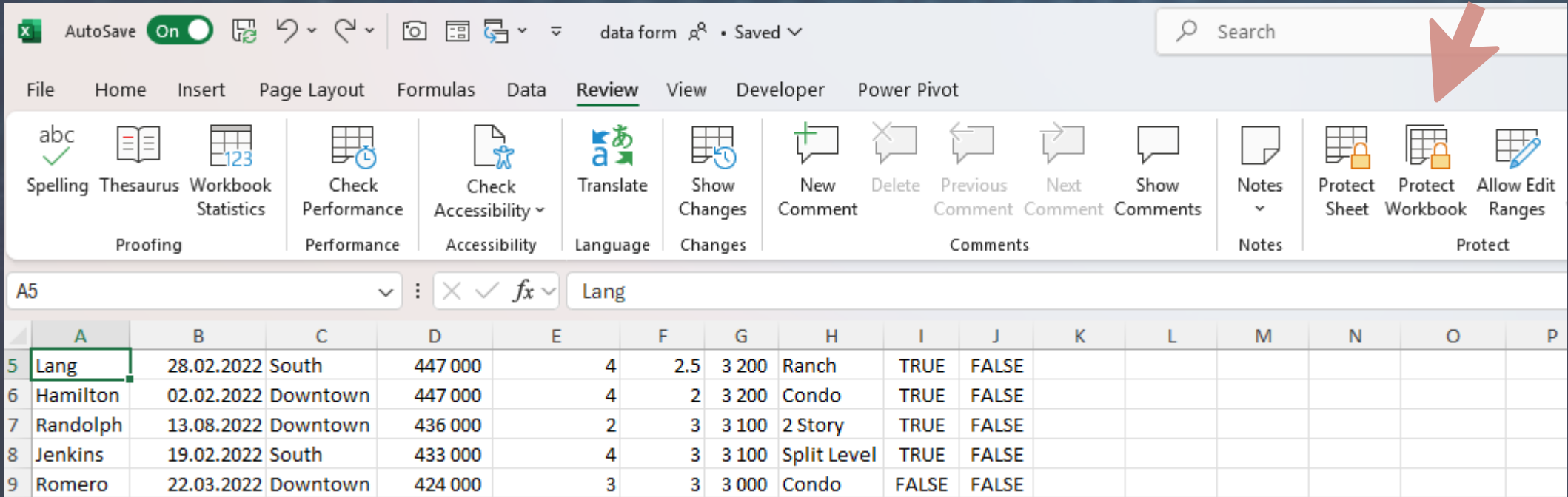
Sheet protection

- Protect sheet will prevent any change in the sheet cells.
- You can set a password on your protection (make sure to remember it).
- If you want to allow some cells to be editable, first select them and press Review > Allow Edit Ranges then click on Review > Protect Sheet.



Protect workbook

- Review > Protect workbook will prevent the sheets structure to be modified.
- You will no longer be able to add, move, copy, delete, rename, hide or unhide sheet. Nor will you be able to edit the tab's color.



The screenshot shows the Microsoft Excel interface with the Review tab selected. A red arrow points to the 'Protect Workbook' button in the Protect group. The interface includes the ribbon tabs (File, Home, Insert, Page Layout, Formulas, Data, Review, View, Developer, Power Pivot), the Quick Start section (AutoSave, Save, Undo, Redo, Recent, Search), and the Review tab ribbon with groups like Proofing, Performance, Accessibility, Language, Changes, Comments, Notes, and Protect. The Protect group contains 'Protect Sheet', 'Protect Workbook', and 'Allow Edit Ranges' buttons. Below the ribbon, the formula bar shows 'A5' and 'Lang'. The worksheet displays a table with 17 columns (A-P) and 5 rows of data.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
5	Lang	28.02.2022	South	447 000	4	2.5	3 200	Ranch	TRUE	FALSE						
6	Hamilton	02.02.2022	Downtown	447 000	4	2	3 200	Condo	TRUE	FALSE						
7	Randolph	13.08.2022	Downtown	436 000	2	3	3 100	2 Story	TRUE	FALSE						
8	Jenkins	19.02.2022	South	433 000	4	3	3 100	Split Level	TRUE	FALSE						
9	Romero	22.03.2022	Downtown	424 000	3	3	3 000	Condo	FALSE	FALSE						

More workbook protection

- Click on File > Info > Protect workbook.
- Encrypt with Password will force a password to be provided in order to open the workbook.

